Torsion of a Circular Shaft

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I. Introduction

1) What are the theoretical assumptions made in the twisting of a circular shaft?

The assumptions made in the twisting of a circular shaft are as follows:-

1) The shaft is straight and has a uniform cross-section throughout its length.
2) The material of the shaft is homogeneous, isotropic (properties are the same in all directions).
3) The material should follow Hooke's law, meaning that the shear stress is proportional to shear strain.
4) Circular cross sections remain circular even after twisting.
5) The cross-sections that are perpendicular to the axis must remain plane even after twisting.

2) For a circular shaft with shear modulus G and radius r, loaded at the ends by twisting moment M_t derive the angle of twist ϕ at a distance z from the fixed end.

From symmetry and the plausible assumption that the extensional strains are zero, we can arrive at the following distribution of strains:

$$\epsilon_r = \epsilon_\theta = \epsilon_z = \gamma_{r\theta} = \gamma_{rz} = 0$$
$$\gamma_{\theta z} = r\frac{d\phi}{dz}$$

Using Hooke's law in cylindrical coordinates, we find that the stress components related to the strain components given by are

$$\sigma_r = \sigma_\theta = \sigma_z = \tau_{r\theta} = \tau_{rz} = 0$$
$$\tau_{\theta z} = G\gamma_{\theta z} = Gr\frac{d\phi}{dz}$$

The only resultant of the stress distribution is, therefore, the moment

$$\int_A r(\tau_{\theta z}\, dA) = M_t$$

Where dA is the element of area and the integral is to be taken over cross sectional area A of the shaft.
If we substitute $\tau_{\theta z}$ into the equilibrium relation, we obtain

$$M_t = G\frac{d\phi}{dz}\int_A r^2 dA = G\frac{d\phi}{dz}I_z$$

where $I_z = \int_A r^2 dA$ is called the *polar moment of inertia* of the cross-sectional area about the axis of the shaft. Rearranging the terms and integrating over the length of the shaft we get the following formula.

$$\phi = \int_0^L \frac{M_t}{GI_z}\, dz$$

3) Does torsion produce homogeneous deformation?
For a circular shaft, torsion does not produce homogeneous deformation, with the assumption that all the extensional strains vanish. ($\epsilon_r = 0, \epsilon_\theta = 0, \epsilon_z = 0$) and from symmetry arguments ($\gamma_{r\theta} = 0 = \gamma_{rz}$)
The only non-zero component of strain is $\gamma_{\theta z}$, which is given by

$$\gamma_{\theta z} = r\frac{d\phi}{dz}$$

We also know that each slice of length Δz deforms in the same way as any other, so we can conclude that $\frac{d\phi}{dz}$ is a constant along a uniform section of shaft subjected to twisting moments at the ends. We call $\frac{d\phi}{dz}$ the twist per unit length. From here, we can see that $\gamma_{\theta z}$ varies with r and for homogeneous deformation the whole body should experience the same strain but this is not true in the case of torsion.
4) What is the polar moment of area?
The polar moment of area basically describes the cylindrical object's (including its segments) resistance to torsional deformation when torque is applied in a plane that is parallel to the cross-sectional area or in a plane that is perpendicular to the object's central axis.
It is usually denoted by I_Z. However, sometimes J or J_Z is also used. It can be represented mathematically with the following formula:

$$I_z = \int_A r^2\, dA$$

5) How to calculate modulus of rigidity from a torsion test?
For a shaft of length L which is loaded at the ends by twisting moments, the total angle of twist between the ends is obtained by integrating the equation below

$$\phi = \int_0^L \frac{M_t}{GI_z}\, dz = \frac{M_t L}{GI_z}$$

Rearranging the terms we can solve for modulus of rigidity (G) by the following equation:-

$$G = \frac{M_t L}{\phi I_z}$$

II. Procedure

1) Switch on the power supply and remove the emergency stop of the machine.
2) Perfectly align the two clamps of the machine and place the specimen on the clamps.
3) Close the safety window of the machine.
4) On the computer, open the required software and open a new file.
5) Check all required details such as the pre-torque, Shear strain rate, alignment angle, upper torque limit, and the unwinding rate.
6) Start the test and wait till the software puts an automatic stop to the test.
7) Unwind and clamp out the specimen and export the collected data to the memory.
8) Repeat the test for other specimens.

III. Observations

In this experiment, we obtained the twisting moment versus angle of twist data for three materials: Aluminium, Brass, and Stainless Steel. The testing machine recorded the data until the specimen fractured (like Brass) or the machine met its preset stopping criterion (such as stainless steel).

The following are the twisted specimens of each material. (The chalk mark on the specimen proves the material has been twisted and by what extent.)

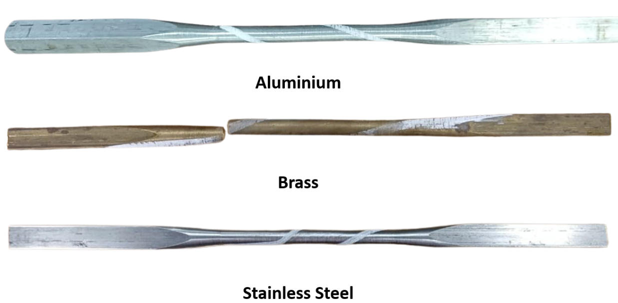

Fig. 1: Twisted material specimens

The plot of the Twisting Moment vs the Angle of Twist, received for the three materials, is as follows.

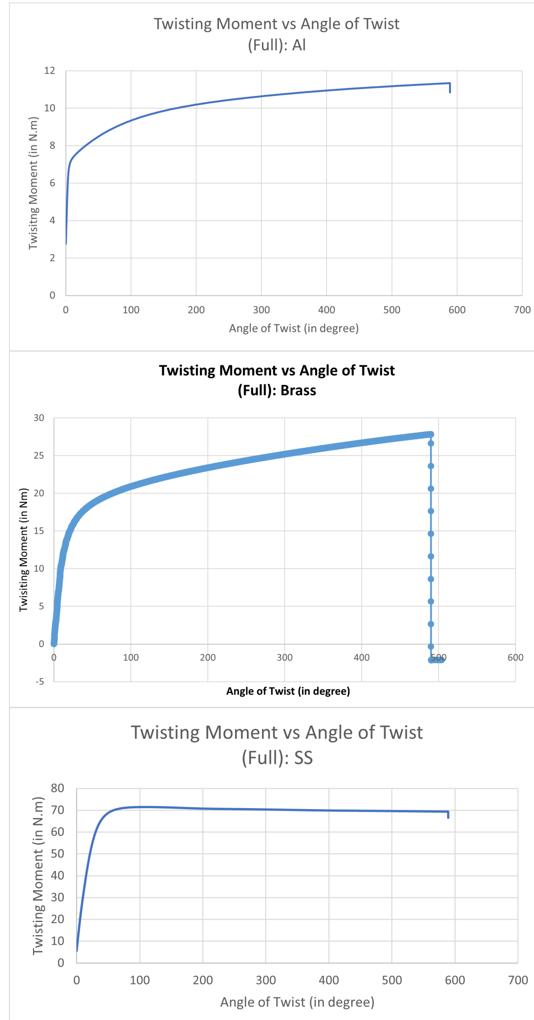

Fig. 2: Full plot of the twisting moment vs the angle of twist

Next, to compare the experiment data with the theory, we chose 3 data points for each material, such that the angles of the twist are small and are within the material's linear (elastic) region.

The plot of the Twisting Moment vs the Angle of Twist for small angles /elastic region (the initial portion of the previous plots) for the materials is as follows. The 3 data points are chosen from this region.

We calculate the twist angle (ϕ) using the applied twisting moment (M_t) by the theory and then compare it with the experimental value. The formula for calculating the twisting angle is below (as derived in the beginning sections).

$$\phi = \frac{M_t L}{GJ}$$

where, L is the length of the specimen (circular part), G is the modulus of rigidity of the material and J is the polar moment of inertia of circular cross-section of specimen.

The three chosen data points for each material and the corresponding calculations using the above formula are summarised

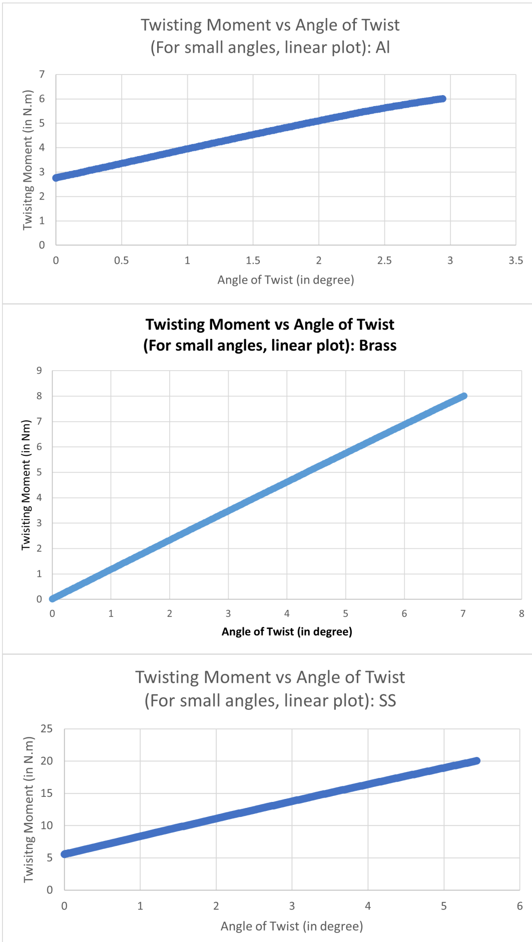

Fig. 3: Elastic region plot of the twisting moment vs the angle of twist

in the following table. (**The link for the Excel workbooks containing raw data and calculations is attached at the end.**)

The modulus of rigidity G, diameter of specimen D and length of specimen L are summarised below for each material:

Aluminium: $G = 27$ GPa, $D = 7$ mm, $L = 7$ cm

Brass: $G = 40$ GPa, $D = 7$ mm, $L = 7$ cm

Stainless Steel: $G = 77.2$ GPa, $D = 7$ mm, $L = 7$ cm

TABLE I: Experimental and Calculated Angles of Twist for Different Materials

Material	**Applied M_t (in Nm)**	**Calculated ϕ (in degree)**	**Observed ϕ (in degree)**
Aluminum	3.00461	1.89	0.208068
	4.50608	2.84	1.47174
	6.00704	3.78	2.94161
Brass	3.00308	1.28	2.582867
	5.00113	2.13	4.33250
	8.00784	3.41	7.01518
Stainless Steel	8.03012	1.76	0.884826
	14.0377	3.09	3.09882
	20.0376	4.41	5.43183

Qualitative Observation: From Figure 2, aluminium and stainless steel exhibit greater ductility than Brass, sustaining large plastic deformation (up to $\sim$590° twist) without fracture, while Brass fractured earlier at $\sim$500°. Figure 1 supports this, showing clean shear failure in the brass specimen due to its ductility, as all metals or alloys are relatively ductile and fail primarily under shear stresses. In contrast, aluminium and steel only stopped deforming because the testing machine reached its limit, not because of fracture. A comparison with a chalk piece shows brittle fracture along a 45° plane, highlighting its weakness under tensile load.

Quantitative Observation: The experimentally recorded twist angles differed significantly from theoretical values due to several factors. Figure 3 shows a linear graph with a different slope ($\frac{GJ}{L}$) than the theoretical, suggesting errors in modulus or dimensions, and a nonzero y-intercept, indicating testing machine inaccuracies.

The detailed explanations and conclusions of the experiment are presented in the following section.

IV. Conclusion

The experiment successfully demonstrated the torsional behavior of brass, aluminum, and stainless steel. The test showed that aluminum and stainless steel are more ductile than brass, maintaining a large twist angle without fracture, whereas brass failed earlier due to shear along a characteristic plane. Overall, the experiment showed elastic and plastic behavior and the shear failure mechanism for metals under torsional loading.

V. Sources of Error

1) Non-proper alignment of the specimen on the machine's loading axis.
2) Small inaccuracies in the specimen dimension would have led to deviation in the theoretical values.
3) The pre-torque(10 Nm) was applied by the machine to perfectly clamp the specimen, before twisting was not considered in the theoretical calculation.

Data Link

https://github.com/MukeshK17/Torsion-Experiment-3.git

VI. References

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